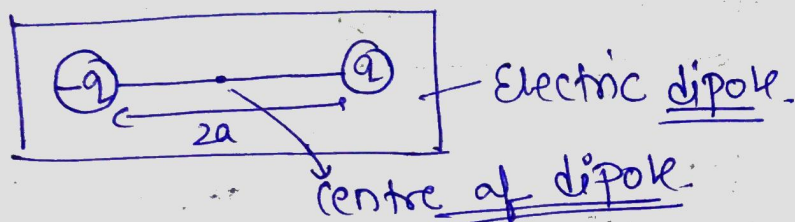


Electric dipole.

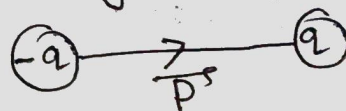
An electric dipole is a pair of equal and opposite point charges q & $-q$, separated by a distance $2a$.



(P) Dipole moment :- product of magnitude of either charges to separation b/w them.

$$P = q \cdot 2a$$

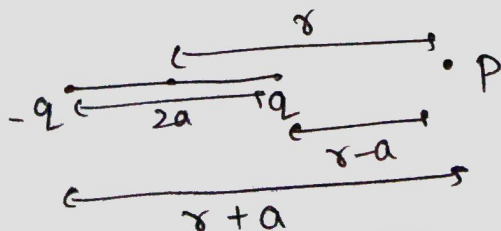
Dipole moment is a vector quantity
Direction $-ve$ to $+ve$.



S.I unit Cm

Electric field due to an electric dipole

① at axial point



$$E_{net} = \frac{2KP}{(r^2 - a^2)^2}$$

② at Equatorial point

if $a \ll r$

$$E_{net} = \frac{2KP}{r^3}$$